LESSON 6.5

Write and Evaluate Algebraic Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an algebraic expression for a real situation and evaluate it for a given value of the variable.

WHY THIS MATTERS Evelyn is helping to plan the sixth-grade field day. The committee will order the items shown for each student participating in field day. They do not yet know how many students will be participating. The shipping cost for all items is a one-time cost of $23.50. What expression can Evelyn use to determine how much the committee will need to purchase the items?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Variable *Spanish: Variable*	A letter that stands for a number that is unknown or can change.	the m in 14m + 35
Algebraic Expression *Spanish: Expresión algebraica*	A math phrase that has at least one letter.	14m + 35
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 14m, the coefficient is 14.
Constant *Spanish: Constante*	A fixed number in an expression or equation that does not change.	the 35 in 14m + 35
Like terms *Spanish: Términos semejantes*	Terms with the same letter raised to the same power, like 2x and 5x.	4x + 2x = 6x (like terms, same variable); but 4x + 2y cannot be combined (different variables)
Substitute *Spanish: Sustituir*	To replace a variable with a given number before working out the value.	For s = 30: 4.25s + 23.50 → 4.25 × 30 + 23.50

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A taxi charges $3 to start plus $2 for each mile. Write an expression for the fare for m miles.

1. Let m = the number of miles.
2. $2 repeats each mile: 2m.
3. The $3 start fee happens once. Join: 2m + 3.

Answer: 2m + 3 dollars

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The field day committee orders one $4.25 item for each student, plus a one-time $23.50 shipping fee. The number of students is unknown.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I let _____ stand for _____ .
- The coefficient _____ shows the amount for each _____ .
- The constant _____ happens only _____ .
- My expression for this situation is _____ .

Word bank variable | algebraic expression | coefficient | constant | term | each | one-time | per | stands for | define | total | unknown

WATCH OUT!

A common mistake

A common mistake in Write Algebraic Expressions is treating 'each' or 'per' as if it means add instead of multiply, and attaching the variable to the wrong number. For 'a $25 setup fee plus $8 per hour, h,' students often write 25 + 8 + h (adding all three numbers) or 25h + 8 (multiplying the flat fee by h) instead of the correct 8h + 25. Since $8 applies per hour, it must multiply the variable (8h), and the flat $25 fee is a constant added only once

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression represents 'the product of 6 and a number n'?

(A) 6n
(B) 6 + n
(C) n − 6
(D) n ÷ 6

STEPS TO TRY
1. Underline the key word 'product.' Every operation has a code word — which operation does 'product' name?
2. Test with plain numbers: the product of 6 and 2 is 12. Which operation turned 6 and 2 into 12?
3. Use that same operation to join 6 and n. Writing a number right next to a variable shows it.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which expression represents '9 less than a number y'?

(A) y − 9
(B) 9 − y
(C) y + 9
(D) 9y

STEPS TO TRY

1. 'Less than' is a tricky phrase — it flips the order of what you write.
2. Try it with numbers: '9 less than 20' means 20 − 9, not 9 − 20. Start with the amount you're taking from.
3. Copy the number pattern: '9 less than 20' → 20 − 9. Use the same order with y in place of 20.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which expression represents 'the sum of a number m and 15'?

(A) m + 15
(B) m − 15
(C) 15m
(D) m ÷ 15

STEPS TO TRY

1. Circle the key word 'sum.' Which operation gives you a sum?
2. Check with numbers: the sum of 2 and 15 is 17. What did you do to 2 and 15 to get 17?
3. Take the operation that 'sum' names and place it between m and 15.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Evaluate 4.25s + 23.50 when s = 10.

(A) $47.75

(B) $66.00

(C) $66.75

(D) $277.50

STEPS TO TRY

1. Replace s with 10.
2. 4.25 × 10 = ?
3. Then add the one-time 23.50.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

EXPLAIN YOUR THINKING

What key words helped you decide which operation to use? How can the same phrase be written differently?

SENTENCE STARTER The phrase '___ ' tells me to use ___ because the key word ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help